

Procedure for Evaluation of Blocaïne

(The open source solution for automation)

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1 General information

To evaluate **Blocaïne**, you only need a single PC running two Python interpreters, one for the Blocaïne **Interpreter** and one for the **Block Editor**.

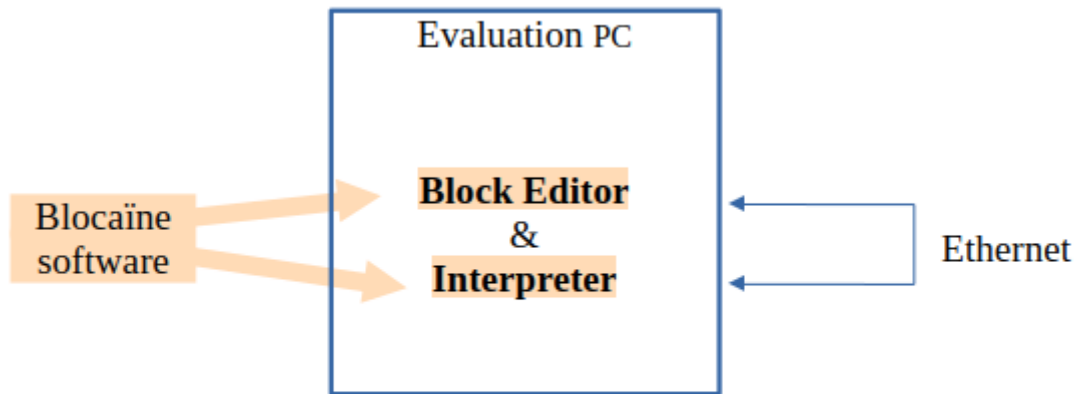


Figure 1 : Evaluation Architecture

2 Prerequisites

You need a computer with at least 100 Mo of free disk and Python3 programming language installed.

2.1 Python3?

Check that Python3 is properly installed on our computer.

On Windows, open a “Command Prompt” (cmd) or a terminal if you are running Linux or macOS, then type the following in the console:

```
python3 --version
```

The response should be like :

```
Python 3.xx.xx
```

If it is not the case, install Python 3 on your computer.

3 Installation of Blocaïne

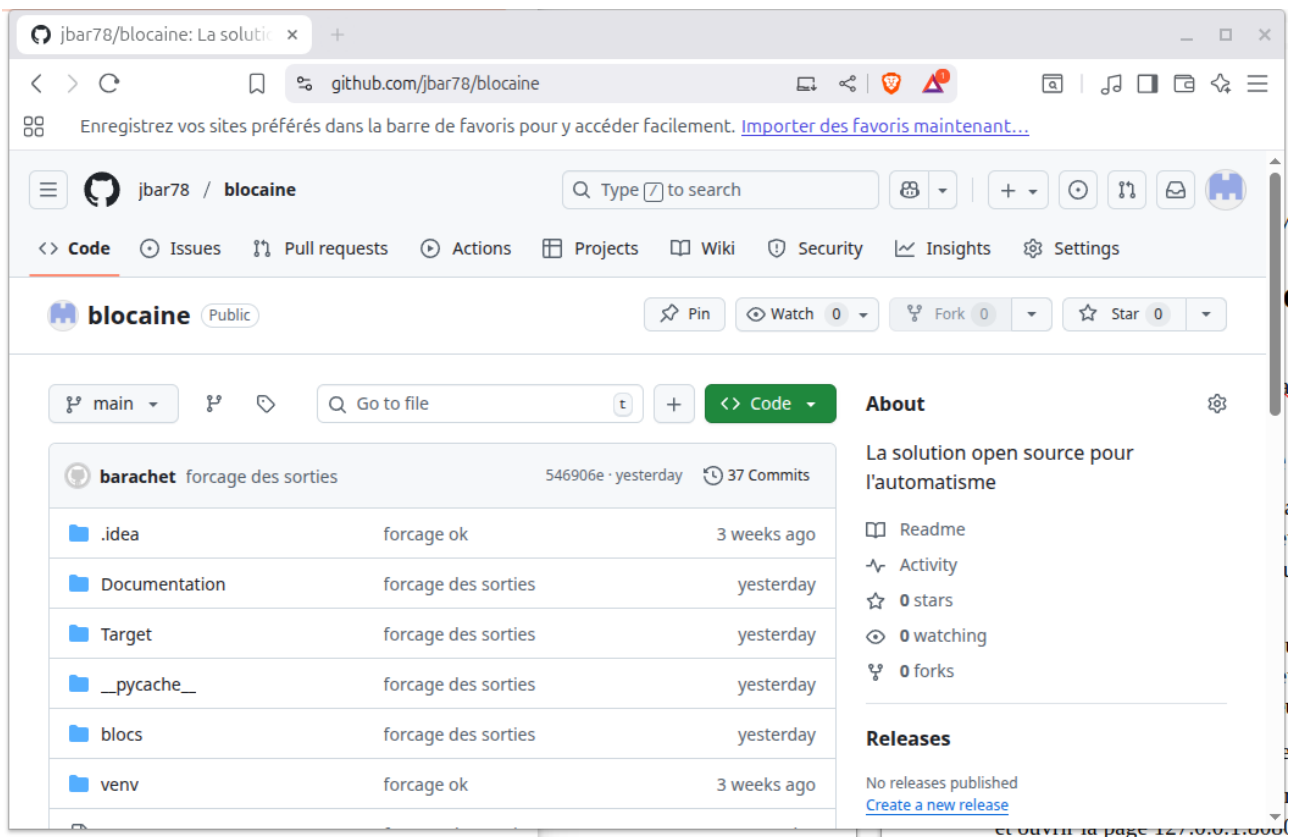
Strictly speaking, there is no real installation process. You simply need to download a ZIP file from GitHub and unzip it. Follow the steps below :

3.1 Dwonloading Blocaïne

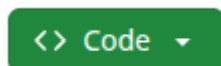
Using your browser, go to:

<https://github.com/jbar78/blocaïne.git>

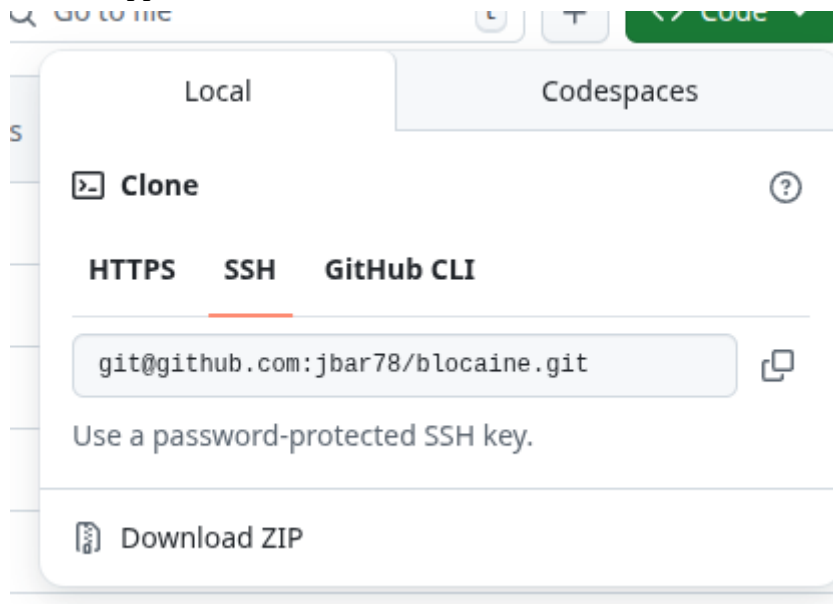
You will arrive on the following page :



Click the main green « code » button.



A popup window will appear. :



Click « Download ZIP », select here you want to save ZIP file and once downloaded, unzip it. you should then have a new directory named «blocaine-main ».

3.2 Installing Python Libraries

The version of Blocaïne you have just downloaded does not require any library (module) other than the standard libraries installed with Python 3, because the ModBus and OPC features have been disabled to make installation easier.

Blocaïne is therefore ready to use.

3.2.1 Using ModBus I/O

If you want to use “ModBus” hard I/O, you must enable this feature by editing the file `PARAM_CONFIG.py` located in the `Target` directory, changing the value of the `PARAM_CONFIG_MODULE_MODBUS` parameter from `False` to `True`, then copy this file into the root directory `blocaïne-main`. You must also install the following library (module): `pymodbus` as follows:

On Windows, type in the console:

```
bash
pip install pymodbus
```

On Ubuntu, type in the console:

```
bash
sudo apt install python3-pymodbus
```

On other Linux distributions, type:

```
bash
python3 -m pip install pymodbus
```

3.2.2 Using the OPC Protocol

If you want the Blocaïne executor to include an OPC UA server so that it can connect to your HMI, you must enable this feature by editing the file `PARAM_CONFIG.py` located in the `Target` directory, changing the value of the `PARAM_CONFIG_MODULE OPCUA` parameter from `False` to `True`, then copy this file into the root directory `blocaïne-main`. You must also install the following library (module): `opcua` as follows:

On Windows, type in the console:

```
bash
pip install OPCUA
```

On Ubuntu, type in the console:

```
bash
sudo apt install python3-OPCUA
```

On other Linux distributions, type:

```
bash  
python3 -m pip install opcua
```

4 Network Configuration

The HTTP and TCP/IP servers within the Blocaïne **interpreter** must use the following IP address: 127.0.0.1

Note: the target address is defined by the parameter `PARAM_TCP_HOST_IP = "127.0.0.1"` contained in the `PARAM_NETWORK.py` files located in the root of the `blocaïne-main` project directory and in the `Target` directory; these two files must be identical.

5 Launching the Blocaïne **Interpreter**

- On Linux: Open the folder “blocaïne-main/Target” in the file explorer, right-click on “launcher.sh”, then select “Run as a program.”
The Python3 interpreter will start, and a terminal window will appear.
- On Windows: Open the folder “blocaïne-main/Target”, double-click “main.py.” If the firewall detects new connections, click Allow.
- On Mac: Run “main.py” inside the “blocaïne-main/Target” directory using the Python3 interpreter.

Once launched, the **Interpreter** is operational. You can now open your web browser and go to <http://127.0.0.1:8080> (port 8080) to navigate through the **Interpreter**’s menu

6 Launching the **Block Editor**

- On Linux: Open the “blocaïne-main” directory in your file explorer, right-click on “launcher.sh,” then select “Run as a program.”
- On Windows: Open “blocaïne-main”, then double-click “main.py.”
- On Mac: Run “main.py” in the “blocaïne-main” project directory using Python3

The Python interpreter will start, and both a terminal and the Block Editor’s graphical interface will appear. The **Block Editor** automatically connects to the Blocaïne **interpreter**.

You can now create your first **user** blocks, upload them, and execute them in the **interpreter**.

7 Uninstalling **Blocaïne**

Since installation only involves downloading and unzipping a ZIP file to obtain the “blocaïne-main” directory, uninstalling Blocaïne simply consists of deleting the ZIP file and the “blocaïne-main” folder.